

Activity 10.1: From Conservation of Energy to Finite Differences

Geophysics — Numerical Heat Flow

Purpose

To understand finite difference heat flow equations as direct consequences of energy conservation applied over small rock volumes.

Physical Concept

Consider a one-dimensional column of rock divided into small cells of thickness Δz . Each cell represents a control volume.

Energy conservation requires:

$$\text{rate of change of thermal energy} = \text{heat in} - \text{heat out} + \text{heat produced}.$$

Tasks

1. Sketch a one-dimensional grid and label temperatures at nodes T_{i-1} , T_i , and T_{i+1} .
2. Write expressions for conductive heat flux entering and leaving a node using Fourier's law.
3. Assuming constant k and uniform grid spacing, show that conservation of energy leads to:

$$\frac{\partial T_i}{\partial t} = \kappa \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta z^2} + \frac{A_i}{\rho c_p}.$$

4. Explain physically what each term in the equation represents.

Key Insight

Finite difference equations are not numerical tricks—they are statements of energy conservation applied locally.